

An Internship Report
Of
DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (CSE)



By

Roll Number of Student:- 2410030132
Name of Student: ARUNDHATI CHAUHAN
Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, ARUNDHATI CHAUHAN, do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ARUNDHATI CHAUHAN

Roll No.: 2410030132

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team and EduSkills Foundation for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Data Science, supported by Siemens. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: ARUNDHATI CHAUHAN

Roll No.: 2410030132

TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate.....	—
3.1 Internship Offer Letter	—
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organization Profile.....	2
4.3 Problem Statement	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used.....	4
4.7 System Architecture.....	5
4.8 Methodology	6
4.9 Expected Outcomes	7
4.10 Certificates of Completion and Communication Proof	8
5. Bibliography/References.....	9

3. INTERNSHIP COMPLETION CERTIFICATE



3.1 Internship Offer Letter



Ref No: C16Y2026M051436470

Internship Offer Letter

Student Name : Arundhati Chauhan
Stream : B.Tech
Branch : B.Tech(Computer Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU69d7b99c553871775745436

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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4. PROJECT DESCRIPTION

4.1 Introduction

This documentation offers a comprehensive review of the 8-week virtual internship completed through the AICTE – EduSkills initiative within the “Data Science Master” track. Conducted from April 2026 to June 2026, the program formed an integral part of the B.Tech CSE degree requirements at IILM University, Greater Noida. The primary focus was bridging theoretical data science principles with practical industry execution using enterprise-standard platforms. Learners were trained to navigate the end-to-end data lifecycle—spanning business use-case evaluation, data engineering, model development, deployment, and platform governance. Upon completing all milestone assessments and the final evaluation, certification was awarded by EduSkills Foundation in collaboration with Siemens.

4.2 Organization Profile

EduSkills Foundation is an ISO 9001:2015 and ISO 20000-1:2018 certified body committed to youth empowerment through technical skill development. Partnering with the All India Council for Technical Education (AICTE), EduSkills delivers scalable virtual internships via the AICTE National Internship Portal. By collaborating with industry leaders, the organization creates practical curricula aimed at boosting engineering graduate employability in AI, Data Science, and cloud platforms.

The “Data Science Master” pathway is backed by Siemens Industry Software and executed on the RapidMiner platform (an enterprise environment integrated via Altair/Siemens). RapidMiner delivers comprehensive infrastructure for ETL processes, predictive modeling, deployment, and administration via RapidMiner Studio and RapidMiner AI Hub.

4.3 Problem Statement

Standard academic programs frequently prioritize theoretical machine learning over end-to-end industry implementations. Engineering students often lack hands-on experience in framing practical business objectives, orchestrating automated data pipelines, and managing deployed models within an enterprise framework. Building the model represents only a fraction of real-world requirements; scalable data engineering and platform governance are equally critical. This internship addressed these industry readiness gaps by providing systematic exposure

across the full lifecycle—from data preprocessing and modeling to operational deployment and enterprise platform administration.

4.4 Project Objectives

- Formulate and translate real-world business challenges into structured ML problem statements.
- Acquire practical expertise in data access, ingestion, transformation, and feature calculation.
- Develop baseline predictive modeling skills using RapidMiner tools.
- Implement advanced data preprocessing and automated workflow techniques.
- Train advanced supervised machine learning algorithms on specialized datasets.
- Package, deploy, and monitor predictive models within simulated production setups.
- Configure, secure, and administer the RapidMiner AI Hub environment.
- Demonstrate technical competence via structured weekly evaluations and a comprehensive final test.

4.5 Scope of the Project

The project encompasses the complete data science pipeline, ranging from initial use-case definition and data engineering to model training, deployment, tracking, and AI Hub administrative configuration. The scope remained within a cloud-based, self-paced simulation environment hosted by EduSkills Foundation without making real-time modifications to live enterprise infrastructure.

4.6 Technologies and Tools Used

- Core Software: RapidMiner Studio, RapidMiner AI Hub, RapidMiner Go
- Data Engineering: Automated workflows for ingestion, cleaning, transformation, and computation
- Machine Learning: Supervised learning models, validation schemes, and performance metrics
- Deployment & Operations: Web-service endpoints and model performance tracking
- Administration: User permissions, resource allocation, and AI Hub governance
- Training Platform: EduSkills LMS for module access and progressive assessments

4.7 System Architecture

The technical workflow implemented throughout the training aligns with the following structured tiers:

- Business Understanding Tier: Converting organizational requirements into measurable AI/ML problems.
- Data Engineering Tier: Ingesting, cleansing, processing, and aggregating datasets in RapidMiner Studio.
- Model Building Tier: Developing and validating analytical models through visual workflow pipelines.
- Deployment Tier: Publishing validated models as operational web services using RapidMiner AI Hub.
- Monitoring Tier: Tracking model metrics, latency, and data drift over time.
- Platform Administration Tier: Managing platform security, compute capacity, user roles, and service health.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and weekly assessments, as summarized in the table below.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Identifying business problems suitable for AI and Machine Learning.
Week 2	Data Engineering Professional	Accessing, loading, transforming, and calculating data.
Week 3	Machine Learning Professional	Building foundational machine learning skills.
Week 4	Data Preparation Master	Mastering advanced data handling and automation.
Week 5	Machine Learning Master	Learning advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploying and monitoring ML models.
Week 7	Platform Administration Master	Configuring and managing RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations, followed by Final Credential Validation / Internship Grade Point Assessment.

Progression required submitting weekly evaluations alongside designated project documentation. Cumulative performance across weekly tasks, documentation submissions, and the final assessment determined the final grade.

4.9 Expected Outcomes

- Competence in mapping business requirements to machine learning workflows.
- Practical ability to execute automated data transformation and engineering pipelines.
- Solid proficiency in building, tuning, and verifying ML models in RapidMiner.
- Hands-on ability to expose models via web endpoints and monitor post-deployment drift.
- Knowledge of managing and securing enterprise analytics infrastructure.
- Improved professional profile backed by industry-recognized credentials supported by Siemens.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter (Ref No. C16Y2026M051436470) and the Internship Completion Certificate (Certificate ID: 4d1aa585ab663cde027e, AICTE Student ID: STU69d7b99c553871775745436) issued by EduSkills Foundation, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Data Science Master, supported by Siemens, are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Foundation, “AICTE–EduSkills Virtual Internship Program – Data Science Master,” Offer Letter, Ref No. C16Y2026M051436470, 2026.
2. EduSkills Foundation, “Certificate of Virtual Internship – Data Science Master,” Certificate ID: 4d1aa585ab663cde027e, 2026.
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4. EduSkills Foundation Portal, internship.eduskillsfoundation.org.
5. RapidMiner Documentation, Altair/Siemens, rapidminer.com.